Differential expression analysis of pancreatic islets using single cell data from HPAP, IIDP and Prodo

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Result summary

Cell type	Control samples	T1D samples	Number genes	Number of latent variables (k)	Number of DEGs up-reg in T1D	Number of DEGs up-reg in control
Acinar	48	12	11,653	3	206	4
Active Stellate	28	2	10,621	4	24	12
Delta	30	5	9,998	3	221	437
Alpha	59	11	13,651	2	21	13
Beta	56	7	13,191	8	596	374
Ductal	38	10	12,151	5	54	32

Abbreviations: DEGs: differentially expressed genes, T1D: type 1 diabetes, up-reg: upregulated

Methods

As the PanKbase single cell map consists of a wide collection of data from multiple tissue sources, isolation centers and independent studies, we used a latent variable analysis to discern biological variation from technical variation as we conducted analyses of differentially expressed genes (DEG). In particular, using DESeq2 (v1.42.1) [1], we compared gene expression profiles between type 1 diabetes (T1D) and control samples for every cell type with more than 20 samples and for each sample with more than 20 non-treated cells; one sample per donor was randomly selected when replicates existed. To minimize potential effects of known and unknown confounding factors, we included known covariates in the DESeq2 model as well as accounted for unknown covariates using the RUVSeq latent variable approach (v1.36.0) [2]. In brief, we used the following multi-step process:

- Step 1: We removed genes from the raw count matrix that had less than 10 reads in fewer than 25% of the samples for that cell type.
- Step 2: We ran a first-pass differential expression analysis using DESeq2 with sex, age, BMI, ethnicity and chemistry as known covariates. The output result was filtered for genes that were not differentially expressed between T1D and control samples which had nominal p values > 0.5. These genes were used as negative control genes for the RUVSeq::RUVg function to estimate latent variables accounting for variation in the data not attributed to disease status.
- Step 3: The latent variables estimated from the RUVseq run were then used as additional covariates (in addition to sex, age, BMI, ethnicity and chemistry) for the second run of DESeq2. To select the number of latent variables *k*, we applied the following algorithm:
    - Step 3.1: We carried out latent variable estimation for a wide range of variable numbers (total tested $k \leq 30$).
    - Step 3.2: Out of all tested variables, we define a *k* named "*k_stop*", which is a *k* that either significantly correlates with diabetes status or is the smallest *k* that does not significantly correlate with any known variables (Spearman correlation, Benjamini-Hochberg adj. p value < 0.05).
    - Step 3.3: From $k = 1$ to $k = k\_stop$, we selected the latest *k* such that it correlates with at least one unique known variable that is not identified by any other *k*.

We applied a 5% false discovery rate (FDR) to identify differentially expressed genes between T1D and control samples for each cell type.

Code availability

All code for the analyses are provided at https://github.com/PanKbase/PanKbase-DEG-analysis.

Citation

[1] M. I. Love, W. Huber, and S. Anders, "Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2," *Genome Biol.*, vol. 15, no. 12, p. 550, Dec. 2014, doi: 10.1186/s13059-014-0550-8.

[2] D. Risso, J. Ngai, T. P. Speed, and S. Dudoit, "Normalization of RNA-seq data using factor analysis of control genes or samples," *Nat. Biotechnol.*, vol. 32, no. 9, pp. 896–902, Sep. 2014, doi: 10.1038/nbt.2931.